A Modular Task-to-Device Integration Framework for Industrial Agents

***Abstract*—Industrial agents must turn task goals into device operations across changing production contexts and heterogeneous interfaces, while preserving evidence of both commands and process responses. The platform connects registered agent harnesses and direct-operation calls to a shared dispatcher. Scene mappings link semantic controls to configured drivers; command receipts remain separate from later process observations. When approval is required, the handler checks that the line, product, recipe, and active run still match the approved context. In simulation, one registered harness completed six sequential tasks across two scenes, each with three admitted candidate settings, synthetic approval, a verified trial receipt, and a run-tagged sample. All six single post-trial probes fell below their target bands. A mock-driver test rejected injected stale contexts before writes. Four simulated deterministic direct-node mission summaries reported final primary values of 32.36 g, 25.30 μm, 99.5 HV, and 3.54 mg/L, each within its scenario-specific target band; no model-selected setpoints were used. These results cover selected configurations; repeatability, comparative optimization performance, human-review effectiveness, and physical-plant performance remain unverified.**

***Index Terms*—Industrial agents, industrial communication, industrial informatics, production systems, semantic data integration.**

I. Introduction

INDUSTRIAL-AGENT deployments translate task-level goals into operations over production signals and equipment. A line-speed adjustment, for example, depends on the active product, recipe, run, applicable limits, and the control node mapped to the line. If that context changes while a proposal is awaiting approval, a formerly admissible request may no longer be appropriate. Device access also needs evidence at two distinct points: what the command interface reported and what the process was later observed to do. Connecting these steps across task and device interfaces is therefore a central industrial integration problem.

Recent work explores end-to-end automation, control-sequence generation, modular factory skills, and robotic-line control [1]–[4]. Other studies address programmable logic controller (PLC) code generation, context-aware control, resource discovery, shop-floor coordination, digital-twin fault handling, and natural-language machine access [5]–[10]. Recent platforms further connect proposals to Asset Administration Shell (AAS) models and signed approval, validate actions against process and instrumentation diagrams (P&IDs), or organize sensor–manufacturing execution system (MES)–enterprise resource planning (ERP) evidence streams [11]–[13]. The cited work covers agent reasoning, production context, and operation constraints.

An integration question remains: how can task engines and direct calls share an operation boundary while accounting for scene-specific devices, production context, and route-specific evidence? This paper contributes a task-to-device integration design that routes registered harness tasks and direct REST calls through a shared named-operation dispatcher. Scene blueprints map semantic signals to configured drivers and define targets and guards, while owning services retain command receipts separately from later process observations.

Fig. 1 summarizes the control path. Task goals and production context inform an operation proposal. Shared checks and route guards precede writes through the device write-control (DCW) subsystem. When approval is required, the handler rechecks the approved context before dispatch. Command receipts and subsequent data-acquisition (DAQ) observations remain distinct: a readback can verify a setpoint, while a later observation describes the process state without, by itself, establishing causation.

Evaluation covers simulator protocol and node access, registered-task execution, mock-driver approval-context checks, and four process missions under separate setups.

Section II reviews related work. Section III presents the architecture and execution flow. Section IV explains the cooperating modules and their device access and evidence mechanisms. Section V describes the simulation protocol and results; Sections VI and VII discuss their implications and conclude.

II. Related Work

Table I summarizes selected plant-facing systems' reported mechanisms and evaluation settings. It supports positioning,

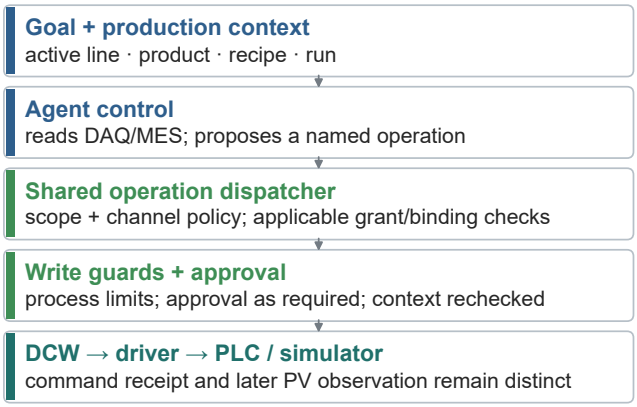


Fig. 1. Agent-control path for a configured industrial task. The agent uses available DAQ and MES context, passes shared operation checks, and routes writes through process guards. Approval and its context recheck apply when required. DCW and configured drivers provide device access; command receipts and later process observations remain separate. Scene configuration supplies targets and guards, while the application provides its optimization policy.

not a matched performance comparison or an exhaustive account of their implementations.

A. Industrial Agents and Plant Integration

Industrial-agent systems address control at several levels. A large language model (LLM) framework for end-to-end automation [1] and control generation for industrial cyber-physical systems [2] complement agent-based supervision [14], PLC code generation, and human–LLM control [5], [6]. Manufacturing studies also explore resource capabilities [7], shop-floor coordination [8], and digital-twin fault handling [9]. Natural-language machine access [10] and real-time LLM control of a robot-operated line [4] extend this work to operator interfaces and robotic-line control.

Xia et al. connect cognitive, digital-twin, and physical-automation layers and organize twin information as state snapshots, event messages, and plan sequences [15]. Other nearby designs expose factory-module operations through the Model Context Protocol (MCP) with state-based tool filtering, constrain manufacturing plans, or compose semantic recipes from OPC Unified Architecture (OPC UA) skills [3], [16], [17]. Several recent systems also combine agent execution with approval, industrial data, or audit records. Graul et al. ground proposals in the Asset Administration Shell (AAS) and route approved actions through OPC UA Methods, with signed approvals and audit logging [11]. Schall separates advisory language-model agents from process and instrumentation diagram (P&ID)-grounded verification and execution, and evaluates structural checks across five simulated process scenarios [12]; its context includes supervisory control and data acquisition (SCADA) and standard operating procedure (SOP) information. Lin and Chen build a governed sensor–MES–ERP digital thread for closed-loop monitoring and control on real manufacturing records [13]. These designs address different parts of the integration problem: asset-grounded approval, action validation, and the organization of industrial evidence. The present paper describes a complementary task-to-device path in which registered tasks and direct calls reach named operation handlers and route-specific records remain with their owning services. The cited studies' reported descriptions do not permit a matched comparison of that entry-path arrangement; Table I therefore presents their mechanisms and evaluation contexts rather than inferring missing capabilities.

The operation checks complement safety-aware decision support [18] and operate alongside the plant's configured process protection.

B. Production Context, Agent Interfaces, and Automation

Industrial integration also depends on abstractions that are not specific to agent architectures. The Model Context Protocol (MCP) specifies interoperable tool discovery and call mechanisms [19]; the operation-level scope, policy, grant, and binding checks discussed here are the platform dispatcher rules, distinct from MCP transport authorization. In parallel, ISA-88 and ISA-95 describe recipes and production operations [20], [21]. Run-to-run methods connect process observations to successive production runs [22]. Human–automation research also formalizes the allocation of information acquisition, analysis, decisions, and actions between people and automation [23]. At the agent level, AgentSpec and CaMeL address tool behavior and capability-based information flow [24], [25]. These standards and agent-level mechanisms describe interfaces for production context, tool interaction, and information flow. The cited reports do not establish whether registered tasks and direct-operation calls share one named-operation boundary with scene-specific checks and service-owned records; this is the integration property examined in the simulator.

III. System Design

The platform separates task execution from operation dispatch while allowing both to use current production context. Shared and route-specific checks govern device routing, and service-owned records distinguish command receipts from measured process responses.

A. Design Rationale and Requirements

Five requirements—scene variability, harness variability, operation authority, outcome observability, and extensibility—shape the design. Scene variability is represented by blueprints and semantic driver mappings; harness variability by registered adapters; operation authority by shared dispatch checks and route-specific guards and admission; outcome observability by service-owned task, receipt, and observation records; and extensibility by registered interfaces for agent engines and device drivers. Tool visibility is separate from write authorization, and actuator writes converge on the shared DCW admission point.

A scene blueprint bundles the process preset, signal inventory, DAQ/DCW driver mappings, and target and guard rules; judged operation records and later observations remain distinct evidence.

During setup, the integrator binds a validated blueprint to the active line–product–recipe–run tuple, maps semantic nodes to drivers, and declares target bands and guards. At runtime, the dispatcher checks current authority before the handler issues a command; the handler returns a receipt, and DAQ records subsequent observations.

B. Layered Architecture

Fig. 2 summarizes the access routes, shared dispatch path, production services, semantic I/O, configured drivers, and service ownership of records. Task and chat routes enter the collaboration band, which contains workspaces, channels, task queues, shared memory, mailboxes, task templates, and harness adapters. A *task route* is an execution path through this runtime; a *harness adapter* connects an agent engine to its task execution interface. Direct REST tool calls and the stdio MCP operation bridge bypass task execution in that band (the upper-right direct-operation entry in Fig. 2) and enter the shared governance block without creating a task or starting a harness turn; MCP management and task methods remain on their service routes. Direct calls resolve current route state at invocation, without an agent-turn prompt or a task record. The governance block projects the operation catalog and applies

TABLE I
REPORTED OPERATION MECHANISMS AND EVALUATIONS IN SELECTED PLANT-FACING STUDIES. EVALUATION SETTINGS AND OUTCOMES ARE SOURCE-SPECIFIC; THIS TABLE DOES NOT COMPARE PERFORMANCE ACROSS STUDIES.

System	Reported operation and evidence structure	Reported evaluation
Xia et al. [15]	Cognitive, twin, and physical layers; twin snapshots, event messages, and plan sequences	Robotic automation and process-simulation cases
Köhle et al. [3]	Injected live state and state-based filtering; MCP-to-OPC UA operations and Message Queuing Telemetry Transport (MQTT) coordination	Nine simulated production challenges
Vieira da Silva et al. [16]	Constrained multi-step plans over MCP manufacturing functions	Laboratory multi-step process
Köhle et al. [17]	Semantic recipes composed from OPC UA skills; virtual commissioning	Unity assembly-workstation proof of concept
Graul et al. [11]	AAS-grounded proposals, signed human approval, PLC-validated OPC UA Methods, and immutable audit log	Two laboratory PoCs; single scripted runs
Schall [12]	P&ID-, SCADA-, and SOP-grounded proposal verification with gated execution	Five simulated scenarios; 43 proposals and 50 LLM runs
Lin and Chen [13]	Governed sensor–MES–ERP digital thread with SOP-, risk-, and capability-aware action gating	25,275 real records; operational metrics
This work	Shared dispatcher and scene-bound DAQ/DCW nodes; separate command and subsequent observation records	Simulator protocol and onboarding checks; six registered-harness tasks; mock approval-context test; four deterministic missions

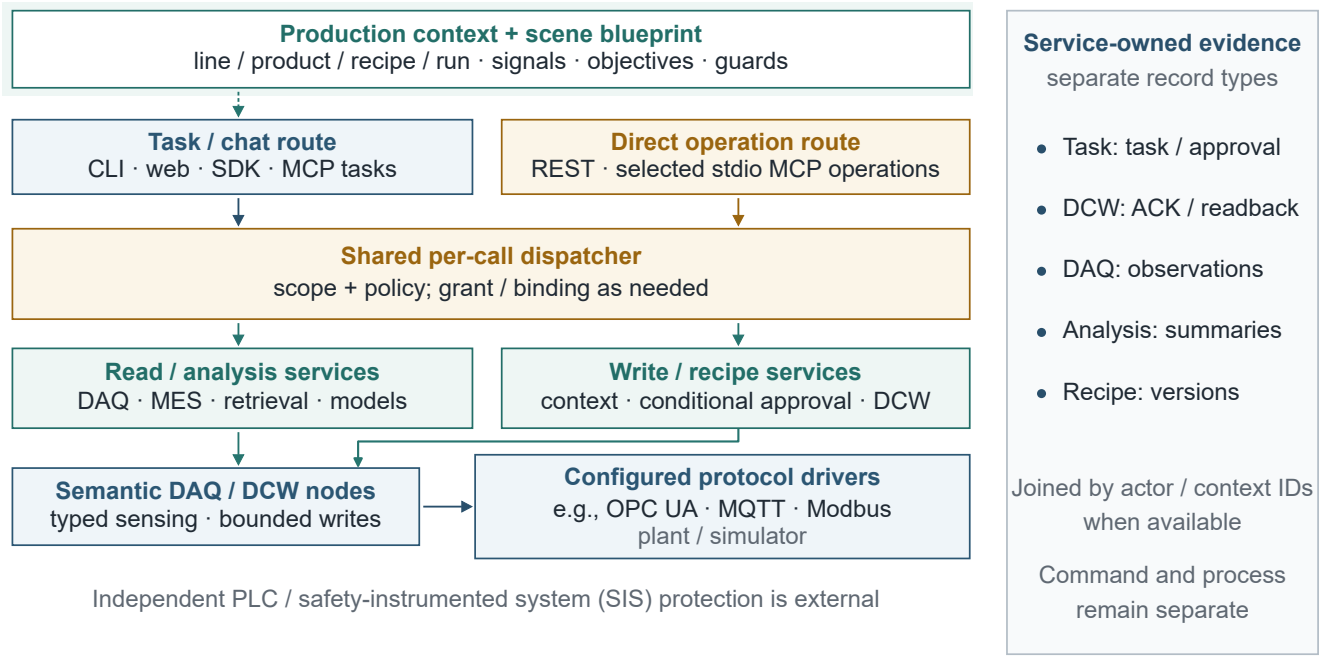


Fig. 2. Architecture and record ownership. Task/chat routes enter the collaboration runtime; direct REST and stdio-MCP calls enter shared dispatch checks without creating a task. One dashed link marks configuration. Write routes apply interval admission and, when configured, approval; recipe dispatch omits the previous-recipe window. The shared dispatcher always checks scope and channel policy, with grant and resource-binding checks applied when required by the operation. Owning services retain distinct records; plant controllers and safety systems retain equipment interlocks.

the per-call checks. Write handlers apply admission limits and request approval when the operation and its configuration require it. The context block supplies the active line–product–recipe–run tuple to the gate and scene maps, objectives, and guards to the handlers. The dashed link from that block in Fig. 2 marks this configuration relationship; the runtime assembles prompt-time context from the same records. The service band, behind a route-handler interface that selects an adapter, provides acquisition, write control, analysis, retrieval, and optional model services. Field I/O resolves semantic nodes to built-in or plugin drivers for a plant or simulator. Services retain task, command, observation, and configuration records separately, linked to actor and production context.

C. Cross-Module Execution Flow

Fig. 3 summarizes a task turn from entry through operation execution, with route-specific evidence retained by owning services. Registered harness adapters receive a scoped tool catalog during setup and supported registry refreshes; the runtime assembles prompt-time production context from current records for each agent turn. For each returned tool call, the shared dispatcher rechecks invocation scope and channel policy and, for industrial operations, current grant and binding conditions before handler selection. The selected handler then applies route-specific guards, active-run checks, approval revalidation, and device-I/O steps as applicable. Command receipts and later process observations remain distinct records.

For an industrial operation u by agent a in channel c, define the shared dispatch condition as

$$\mathcal{A}(a,c,u) = \bigwedge_{X \in \mathcal{C}(u)} X(a,c,u), \tag{1}$$

where S, P, G, and B denote invocation scope, current channel policy, applicable industrial grant, and the agent's binding to referenced resources, respectively. $\mathcal{C}(u) \subseteq \{S,P,G,B\}$ is the set of predicates declared for operation u; it always contains S and P. The grant predicate G is included when the resolved channel, target line, and owner/grant policy require it; explicit exemptions omit it. The binding predicate B is included when the operation declares resource references. A declared no-reference operation omits B. Each included check must pass before handler selection. The policy predicate includes channel tool selection, profile, and duty-separation checks where configured. Once these predicates pass, the selected route applies its guards, active-run checks, admission limits, and any required approval. When approval is required, the handler revalidates its context before device dispatch.

For direct REST invocation, the service establishes a and c before applying these checks: an agent token must match the requested agent identifier, or an authenticated user must be the channel owner or an administrator. The manager loads the channel-agent record and rechecks scope and policy; industrial operations then pass the shared grant and binding checks. Handlers resolve live production state from request identifiers and owning services.

Fig. 4 gives reference pseudocode summarizing the registered-harness task path. Direct REST tool invocation bypasses the task runtime and harness but enters the same shared dispatcher. Route-specific timing, approval, and evidence rules are detailed in Section IV.

IV. Module Functions and Integration

The architecture in Section III assembles task context, dispatches operations, and connects device access to separate records of commands and observations. The following subsections explain the handoffs between these functions.

A. Task, Harness, and Context

Task and chat entry points create or join a workspace channel. The task runtime manages membership, task state, and shared execution context; a harness registry adapts each supported agent engine to a common execution interface. At each agent turn, the prompt builder assembles the task, active scene description, production context, channel profile, agent binding, and retrieved memory. This keeps engine-specific interaction behind the harness interface while preserving the current line and operation context for the agent.

Direct REST and selected stdio-MCP operation calls reach the same dispatcher without a task or harness turn. REST authentication and identity resolution follow Section III-C; each handler reads live route state at invocation. Handlers receive the resolved actor, channel, and operation arguments through the same interface for either entry route; records are keyed by the identifiers supplied in each request. A registered task additionally receives prompt context and a scoped tool catalog. Catalog visibility alone does not grant write authority: the dispatcher checks authority at invocation.

B. Shared Dispatch and Device Admission

The dispatcher is the shared operation boundary. As specified by (1), it checks invocation scope and channel policy for every call; grant is skipped only for a grant-exempt industrial tool, and binding applies when the operation declares resource references. Applicable checks precede handler selection. Handlers revalidate route-specific state at execution time; the dispatcher applies grant checks when required. Scene blueprints define device and signal maps, objectives, and process guards. Capability profiles and tool policies narrow available operations, while bindings constrain accessible nodes, controls, and recipes. Optional DAQ, MES, analysis, retrieval, optimization, and digital-twin services are invoked through registered handlers. The application supplies its optimization strategy while the framework coordinates operation execution.

Actuator writes converge on the shared DCW admission point. For an online node write, the admissible value interval is

$$\mathcal{I}_n^{\mathrm{online}}(t) = \mathcal{H}_n \cap \mathcal{L}_n^{\mathrm{base}} \cap \mathcal{L}_n^{\mathrm{prod}}(t) \cap \mathcal{L}_n^{\mathrm{rec}}(t), \tag{2}$$

where $\mathcal{H}_n$ is the hard engineering range and $\mathcal{L}_n^{\mathrm{base}}$, $\mathcal{L}_n^{\mathrm{prod}}$, and $\mathcal{L}_n^{\mathrm{rec}}$ are the baseline, active-product, and active-recipe limits; the $\mathcal{L}$ limits here are deliberately named apart from the policy and binding predicates of (1). Missing optional limits impose no restriction; an empty intersection rejects the request and identifies the excluding layer. Recipe dispatch validates the replacement recipe against the hard, baseline, and product limits rather than the previous recipe window. Step, interval, and hold-window checks also constrain each route. These software checks bound commands alongside independent programmable-logic-controller and safety-instrumented protection.

C. Approval, Execution, and Evidence

Routes add timing checks and, when required by operation and configuration, approval checks. Online node writes enforce a 60 s minimum spacing; recipe trials use a per-line cadence (300 s by default), and recipe intervals are anchored to approval. Manual approval binds an actor and proposed operation to candidate values, applicable limits, expiry, and

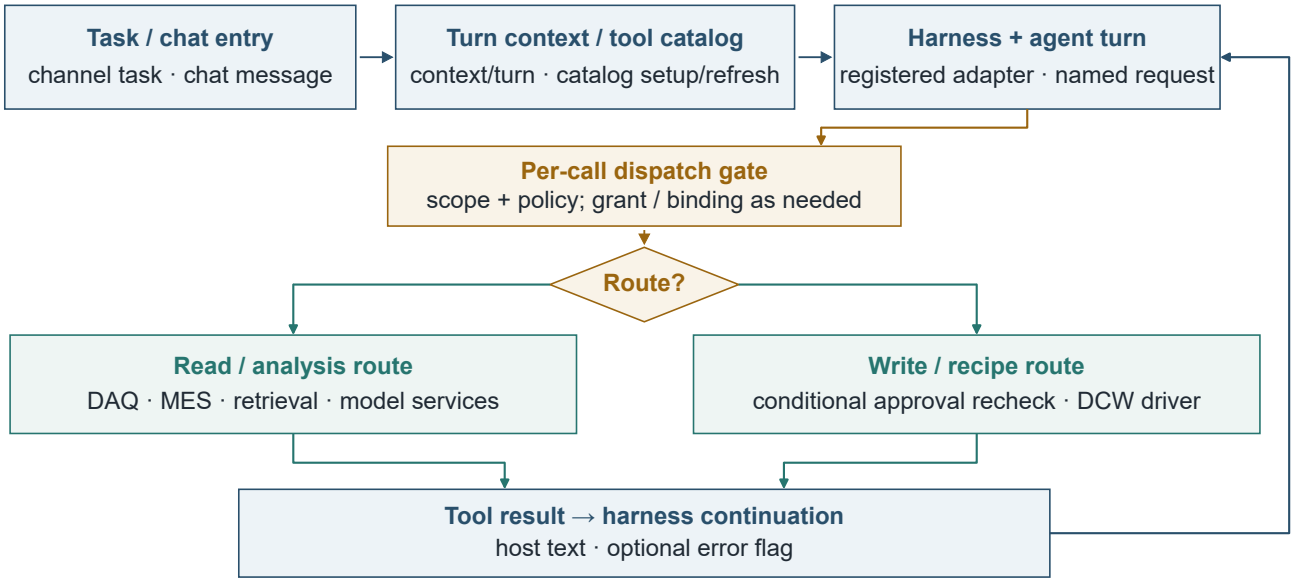


Fig. 3. Registered-harness task flow. The runtime invokes the adapter with a scoped catalog. Scope and channel policy are checked for every call; grant applies to non-exempt industrial operations, and binding applies when resource references are declared. Failed checks return tool errors. The read path retrieves data. The write path applies route guards, conditional approval and its context revalidation, DCW admission, and receipt grading; a later process observation is separate. Results return to the harness. Device-level interlocks remain with the configured plant-control layer.

Require: task q, channel c, bound agent a, registered scoped catalog T
Ensure: task result; any retained evidence remains with its owning service
$x_t \leftarrow \textsc{ResolvePromptContext}(c, q)$
$m \leftarrow \textsc{RetrieveMemory}(q)$
$p \leftarrow \textsc{AssemblePrompt}(q, x_t, m)$
$h \leftarrow \textsc{InvokeRegisteredHarness}(a, p, T)$
for all tool request u returned by h **do**
  $g \leftarrow \textsc{CheckScopeAndChannelPolicy}(a, c, u)$
  if not g **then**
    $\textsc{EmitToolErrorToHarness}$(h,u); **continue**
  end if
  if u is an industrial operation **then**
    $v \leftarrow \textsc{CheckApplicableGrantBinding}(a, u)$
    if not v **then**
      $\textsc{EmitToolErrorToHarness}$(h,u)
      continue
    end if
  end if
  $k \leftarrow \textsc{SelectHandler}(u)$
  $r \leftarrow \textsc{ExecuteRouteHandler}(k, a, c, u)$ ▷ handler resolves live state and may persist route-specific records
  $\textsc{EmitDeclaredToolResultToHarness}$(h,r)
end for
return task state and tool results

Fig. 4. Reference pseudocode for the registered-harness task path. Direct operation calls use the shared dispatcher from their own entry point. Each route persists records through its owning service and returns its declared response to its caller; task-route results return to the harness.

the active line–product–recipe–run tuple. After approval, the handler rechecks authority and compares the current tuple with the approved tuple before entering DCW dispatch. Single-parameter waits expire after 180 s by default; recipe-dispatch approvals use a 30-minute window. A changed recipe definition requires a new `recipe_apply` proposal.

Fig. 5(a) illustrates the four-layer value interval in Eq. (2) using an illustrative rotational-speed configuration in revolutions per minute (rpm), taken from a separate archived admission-limit drill rather than the four process missions. Panel (b) shows a symbolic approval followed by authority and context rechecks; its values illustrate the approval sequence.

A `recipe_trial` illustrates the operation sequence: the handler checks the bound recipe, active run, candidate parameters, and route timing; if required, it requests approval; after revalidation, it submits a bounded batch through the configured driver and returns a per-parameter receipt. The caller obtains later DAQ observations separately and may then update, apply, or roll back a recipe. A trial neither saves nor applies a recipe version. The driver registry maps semantic controls to configured protocol drivers, keeping protocol details out of agent operation arguments.

For a concrete illustration, consider a simulated coating line whose semantic line-speed control is mapped to a Modbus TCP write node. A request to change the setpoint from 37 to 39 m/min is inside a 28–48 m/min recipe window; the remaining applicable limits and timing checks must also pass. If the configured policy requires approval, the proposal binds the candidate and active production context, and the handler rechecks that context before dispatch. The driver then returns a graded receipt. A later acquisition sample is stored separately and can be associated with the operation when the route provides the necessary context and timestamps. This example

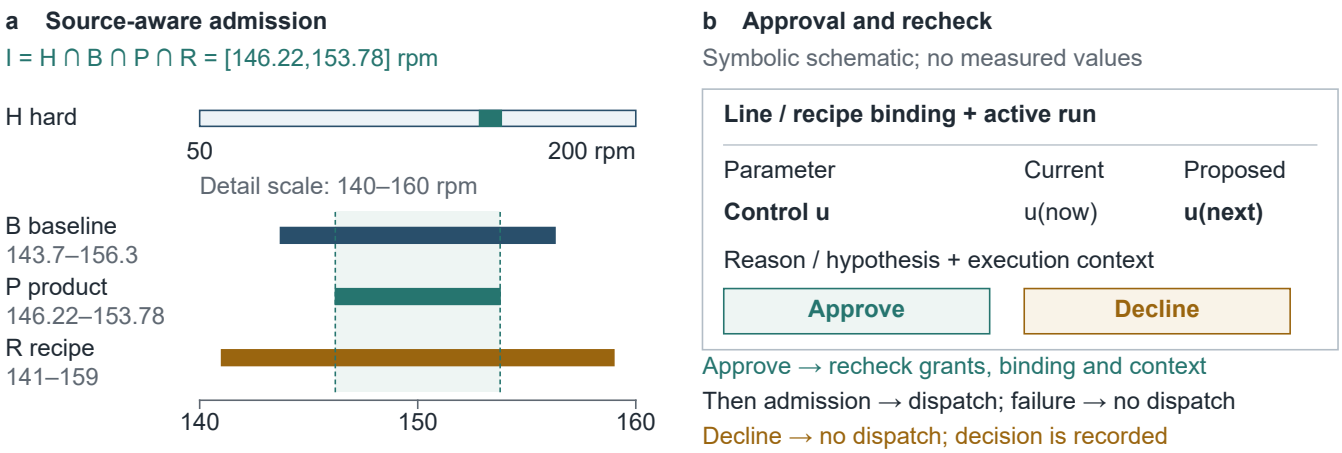


Fig. 5. Admission and approval boundaries. (a) An illustrative rotational-speed configuration in revolutions per minute (rpm): online node interventions use the four-layer interval of Eq. (2), whereas recipe dispatch omits the previous recipe window. (b) Symbolic human approval binds a proposed action and its context tuple; execution rechecks the active line–product–recipe–run context, authority, and applicable limits before device dispatch.

illustrates the handoffs, not a recorded human decision or a particular process response; the simulator onboarding replay in Section V used test-only automatic approval.

Receipts and observations answer different questions. A command result is graded as readback-verified, transport-acknowledged, or unverified; a readback mismatch can trigger up to three reads at 2 s intervals. Recipe batches retain per-parameter verification outcomes. A readback-verified result records that the returned value matched the request; a transport acknowledgement alone does not establish that the device applied it, and an unverified result does not establish a matching readback. DAQ stores subsequent timestamped process observations, while recipe history, approvals, and task records remain with their owning services. Identifiers and timestamps can link these records when available, but a later observation alone does not prove that the command caused the measured change. Section V evaluates selected parts of this evidence chain under the stated test conditions.

V. Experimental Evaluation

The evaluation examines selected parts of the task-to-device integration path using local simulators and a mock driver. A registered-harness test examines task execution and its retained operation records; Fig. 6 shows the associated process probes. Other checks examine protocol and node access, direct-operation records, and approval-context rejection. Separately, four deterministic direct-node missions illustrate how requests and observations are recorded across different process scenes. They cover injection molding, biaxially oriented polyethylene terephthalate (BOPET) film production, continuous annealing, and anaerobic/anoxic/oxic wastewater treatment. Table III summarizes their primary outcomes; their trajectories appear in Fig. S2 of the supplementary evidence bundle. The missions and harness tasks address different evaluation questions and use different setups; neither is a control or repetition of the other.

A. Test Design and Evidence Rules

All process experiments use local simulators, not physical equipment. The four missions use deterministic scenario-specific routines, a fixed random seed, and the direct-node interface; no language model selects their setpoints. In the available benchmark source, an automatic keep judgment follows a write and precedes the next observation, except in BOPET, where it follows the response. Annealing and wastewater logs also record later reversion decisions. These are software events, not human pre-approval. The full executed runtime snapshot was not retained, so the records cannot be replayed as a complete trace.

Each mission contributes one trajectory, with its initial observation as a within-trajectory reference rather than an independent control. A requested setpoint is an action, not a device readback; subsequent process values are observations. The primary outcome is the mission-reported final value relative to its target band. Intermediate observations are not independent repetitions.

B. Protocol and Node Integration

The protocol test configured simulator endpoints using Modbus TCP, Modbus RTU, OPC UA, MQTT, and HTTP; its contemporaneous report records successful connectivity for each. In a Modbus TCP check, seven samples were collected and the simulated process reached within 3 °C of a requested 182 °C setpoint after 24 s. In a separate fault case, a 185 °C write was read back while the process value remained frozen at 150 °C; monitoring raised an alarm within 2 s. A reported link-fault exercise detected loss of connection and recovered after simulator restart. The complete per-sample protocol trace was not retained, so these results are limited to the contemporaneous summary.

The BOPET line had nine devices and 49 signals: 30 writable setpoints and 19 measurements. Its four requests addressed three control nodes. A separate onboarding replay configured one simulated Modbus TCP device with four write and seven acquisition nodes, together with scalar, vector, table,

and image forms from a manufacturing execution system (MES) simulator. All seven acquisition nodes yielded samples, four invalid writes were rejected, and four legal writes were confirmed by readback; all 23 verifier checks passed. A test-only automatic approval allowed a 37-to-39 m/min line-speed change, confirmed by the simulator. This was one integration instance. The source record does not fully identify the build, and the MES image used a local storage fallback.

C. Registered-Harness Tasks and Direct Operations

A registered task harness (an open-model protocol engine operating over stdio JSON-RPC) configured with `glm-5.3-flash` executed six sequential tasks across two simulated scenes. Each task supplied a current control read, process measurement, target band, and three candidate settings admitted by the dispatch predicates of (1) and the interval admission of (2); band attainment itself was not pre-verified. Application records associate the selected candidate with a registered-harness tool proposal; independent model-provider request and response logs were not retained. With synthetic administrator approval, all six tasks completed and each retained an approved proposal, a recipe-trial record, a 1/1 verified device receipt, and a run-tagged sample. The runner initiated a 20 s delay after each receipt/readback checkpoint before probing the driver. Fig. 6 pairs each pre-task process measurement with its single post-trial probe. In the first film task, the pre-task value was 55.82 μm, above the 48–52 μm target band, while the probe was 45.69 μm, below it. The other film probes were 46.69 and 45.69 μm; annealing probes were 679.60, 677.70, and 678.10 °C against a 700–760 °C target. Thus, despite task completion and verified receipts, all six probes were below target. The three tasks per scene shared one simulator instance, and samples used a local SQLite fallback because the isolated online queue and time-series service were unavailable. These paired checkpoints do not establish causation, convergence, independent replication, or normal online acquisition.

Separate checks without model-generated actions exercised a fixed-call registered task and a direct REST operation in each scene, using synthetic approval and isolated environments. Each returned a verified trial receipt and a run-tagged sample; the task route also retained a task–approval–receipt–sample link, but task completion was not asserted. At 20 s, film samples were both 46.37 μm; annealing samples were 674.80 and 675.20 °C, respectively, all below the corresponding target bands. The samples were explicit driver probes stored using a local database fallback with online DAQ disabled. The two checks document which records each route retained. They used separate environments and did not trace both calls to the same handler; neither adds a four-mission trajectory.

D. Approval Context and Dispatch

An independent component test exercised human-in-the-loop (HITL) approval using simulated decisions and a mock driver. A proposal binds the line, product, recipe, and active run to the approval context; after approval the handler compares that context with the current one before dispatch. The test changed each of the four fields for recipe application, trial, and rollback: all injected stale contexts were refused before mock writes. A further injected change between authorization revalidation and the final check was also refused. An unchanged approved context produced the expected mock writes, whereas a rejected proposal produced none.

To examine the contribution of the context comparison, a test-only ablation removed that comparison while retaining the other checks. Six product or run changes then reached the mock driver and each produced two writes and an execution record. The remaining six line or recipe changes were still blocked by other checks. Thus the comparison supplied the tested product/run protection in this component setup; the result is not a measurement of operator effectiveness, human decision time, physical safety, or behavior during network and device I/O. Table II summarizes the integration and approval checks. Table III reports the separate process-mission outcomes.

TABLE II
DISTINCT, NONPOOLED SOFTWARE CHECKS: FIVE PROTOCOL TYPES, ONE SIMULATOR REPLAY, SIX HARNESS TASKS, 13 MOCK-DRIVER APPROVAL CASES, AND 12 MOCK-DRIVER ABLATION CASES.

Check	Observed outcome
Protocol access	Five protocol connections succeeded
Node onboarding	All seven acquisition nodes sampled; all invalid writes rejected; all legal writes read back
Registered harness	Six tasks completed with verified receipts; all six probes outside target
Approval context	12 stale-context changes and one revalidation-race case refused before mock writes
Without context comparison	Six product or run changes admitted; all line or recipe changes still refused

TABLE III
PRIMARY PROCESS VALUES IN THE FOUR SIMULATED MISSIONS. EACH ROW IS ONE TRAJECTORY; THE FINAL VALUE IS THE MISSION-REPORTED VALUE.

Scenario	Target band	Initial → final
Injection	Part weight: 32.15–32.85 g	31.296 → 32.36 g
Film	Thickness: 24.3–25.7 μm	28.03 → 25.30 μm
Annealing	Hardness: 89–101 HV	134.9 → 99.5 HV
Wastewater	Dissolved oxygen: 2.8–4.0 mg/L	0.194 → 3.54 mg/L

E. Direct-Node Missions: Injection Molding and Film

The direct-node film mission uses a different thickness target from the registered-harness tasks. The injection-molding mission targeted 32.5 ± 0.35 g. Starting at 31.296 g, the routine requested holding pressure from 45 to 57.73 bar and holding time from 6 to 10.56 s; observations rose to 31.93 and 32.09 g. A further pressure request to 62 bar preceded the final 32.36 g observation within the target band. This single sequence does not isolate the effect of each parameter.

The BOPET mission targeted 25.0 ± 0.7 μm. From an initial 28.03 μm, four requests changed casting speed, fast-roll speed, rail width, and casting speed again; successive observations were 26.48, 26.06, 25.78, and 25.30 μm. The final value was within the band, while the first three remained above it. Reported melt temperatures were 291.3–291.8 °C, within the

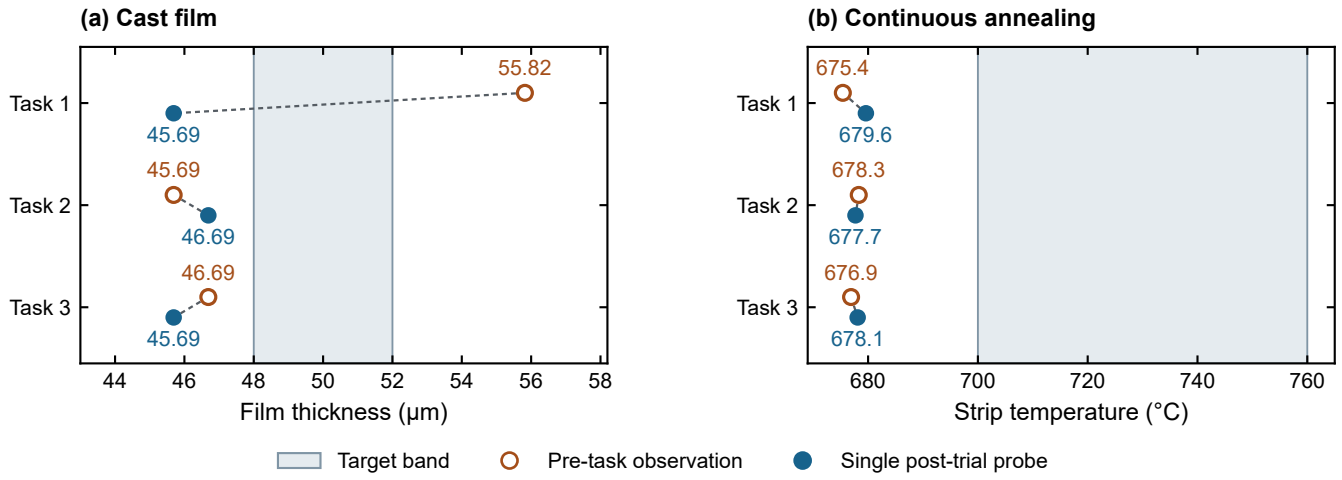


Fig. 6. Pre-task observations and single post-trial probes for six sequential registered-harness tasks in two simulated scenes. Shading denotes the target band; task numbers indicate order within each scene. Dashed connectors pair observations within a task, and vertical offsets separate markers without changing their measured values. The connectors do not represent a time series. The runner waited 20 s after receipt/readback before each probe. Tasks in each scene shared a simulator instance; synthetic approval and a local SQLite fallback were used.

configured 268–300 °C window. These observations describe one simulated sequence.

F. Direct-Node Missions: Annealing and Wastewater

The direct-node annealing mission measures hardness, whereas the registered-harness tasks record strip temperature (Fig. 6). The annealing mission combined a 95 ± 6 HV hardness target with a line-speed increase. From 134.9 HV, a zone-2 temperature request from 710 to 850 °C preceded a structured observation of 99.18 HV; a speed request from 140 to 152 m/min preceded 99.74 HV. A 164 m/min candidate was reverted after event text reported 101.0 HV, at the upper target boundary with no remaining hardness margin. The summary separately reported final hardness of 99.5 HV and retained speed of 152 m/min, 8.57% above the initial setting, not a measured throughput gain. A zone-3 read failed before the zone-2 action. Final tensile-strength and surface-defect measurements were not archived, so the record cannot establish that all quality constraints were met.

The wastewater mission targeted dissolved oxygen (DO) at 3.4 ± 0.6 mg/L. Five blower-frequency increases from 26 to 34.1 Hz accompanied a rise from 0.194 to 3.567 mg/L in the structured aeration sequence. Subsequent alkali and coagulant dosing addressed pH and phosphorus; observed total phosphorus fell from 0.662 to 0.5507, 0.4607, and 0.413 mg/L. A later blower reduction was reverted after insufficient margin. The summary separately reported final DO of 3.54 mg/L and a simulator-defined cost index changing from 182.6 to 180.2, not a measured operating cost. The complete final effluent vector was not archived, so these values do not verify all discharge constraints.

G. Evidence Synthesis

The six registered-harness tasks retained operation records, and each single post-trial probe fell below its target band. Separately, the four direct-node missions reported final primary values within their respective target bands. Neither evaluation establishes repeatability, causal effects of individual actions, comparative performance, human-approval effectiveness, or transfer to a physical plant.

VI. Discussion

A. Interpretation

Protocol connectivity and the onboarding replay test different parts of device integration: communication with configured simulator endpoints and access to declared acquisition and control nodes. They do not show that arbitrary equipment can be integrated without adaptation. The entry-path checks document which records were retained for each tested call. Because the calls used separate environments and no handler-level trace was retained, they cannot establish route equivalence or confirm that both reached the same handler. In the approval-context test, stale product and run identities reached mock writes when tuple comparison was removed and were rejected when it was enabled. This isolates the comparison's contribution within that setup.

The six registered-harness tasks and four direct-node missions answer different questions. The harness tasks link approvals and receipts to single post-trial probes; the missions record one process trajectory per scenario. They use separate setups, including different film target bands and different annealing measures, so mission endpoints do not benchmark the harness probes. Across the missions, requested actions precede measured responses, including a reverted annealing candidate. In wastewater treatment, the structured aeration sequence and final mission summary also refer to different observations. These records describe trajectories, not isolated effects of individual commands.

The missions also illustrate the role of scene-specific configuration. Part weight, film thickness, hardness, and dissolved oxygen have different units, target bands, and avail-

able actuators. The operation–response structure accommodates these differences, but the trajectories do not identify which architectural component caused an outcome. Related approaches likewise connect industrial operations to semantic skills, equipment context, or process validation [11], [12], [17]. The present study complements those designs by describing a shared operation path and reporting responses in four simulated scenes.

B. Scope and Limitations

The optimization evidence comprises one deterministic simulated trajectory per scene, without independent replicates or a matched alternative controller. It cannot establish statistical repeatability, superiority to another platform, or the effect of individual controls. The missions use direct-node operations and do not test the full range of entry routes or language-model decisions described by the architecture. Final tensile-strength and surface-defect readings for annealing and the complete final wastewater effluent vector are missing, limiting conclusions about secondary constraints. The process model and simulator configuration also limit transfer to physical production.

The mission records show automated judgments, not human pre-approval. The approval-context test uses simulated decisions and mock drivers; it tests a software condition, not operator performance or the effectiveness of human review [26], [27]. Software operation restrictions are not a complete security or safety assessment [28]–[30]; independent plant protection remains necessary [31]. Future work should repeat missions across initial conditions, preserve all terminal quality measurements, compare matched controllers and entry paths, and evaluate the system with suitable physical equipment and operator oversight.

This modular separation is a property of the implemented software structure. The current evidence does not measure whether it reduces engineering effort, operating cost, or reliability; those outcomes require matched deployments.

VII. Conclusion

The platform links industrial task context to scene-specific device operations through shared dispatch, configured drivers, and separate command and observation records. The evaluation covered simulator protocol and node access, registered-harness tasks, mock-driver approval checks, and four direct-node missions. The four mission endpoints were within their primary target bands. Separately, all six single probes after harness tasks remained below their bands despite verified receipts. These simulator and mock-driver results do not establish repeatability, comparative benefit, human-review effectiveness, or performance on a physical plant.

Data Availability: The historical Supplementary File S1 contains route and integration-check records and the onboarding replay log; it does not contain the current four-mission traces. A separate supplementary evidence bundle contains pseudonymized mission logs and selected registered-harness records. The manuscript source, vector figures, and figure-source data accompany the submission package. No physical-plant data are included.

References

[1] Y. Xia, N. Jazdi, J. Zhang, C. Shah, and M. Weyrich, "Control industrial automation system with large language model agents," in *Proc. IEEE 30th Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205539.

[2] K.-C. Huang, H.-T. Lin, C.-H. Lin, and M.-T. Huang, "Large language model integration in industrial cyber-physical systems: A multi-modal architecture for automated control generation," in *Proc. Int. Conf. on Control, Automation and Diagnosis (ICCAD)*, 2026, pp. 1–6, doi: 10.1109/ICCAD69956.2026.11643124.

[3] K. Köhle, D. Anicic, T. A. Runkler, and R. Graf, "LLM-driven multi-agent control for skill-based smart manufacturing," in *Proc. IEEE 31st Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2026, pp. 1–8, doi: 10.1109/ETFA68377.2026.11725257.

[4] M. Waseem, K. Bhatta, C. Li, and Q. Chang, "Pretrained LLMs as real-time controllers for robot operated serial production line," *Mach. Learn.: Eng.*, vol. 2, no. 1, 2026, art. no. 015006; doi: 10.1088/3049-4761/ae4ee8.

[5] Z. Liu *et al.*, "Agents4PLC: Automating closed-loop PLC code generation and verification in industrial control systems using LLM-based agents," *IEEE Trans. Softw. Eng.*, vol. 52, no. 5, pp. 1672–1687, 2026, doi: 10.1109/TSE.2026.3667895.

[6] R. Wu, J. Ai, and T. Li, "InstructMPC: A human-LLM-in-the-loop framework for context-aware control," in *Proc. IEEE 64th Conf. on Decision and Control (CDC)*, 2025, pp. 172–179, doi: 10.1109/CDC57313.2025.11312156.

[7] J. Lim and I. Kovalenko, "A large language model-enabled control architecture for dynamic resource capability exploration in multi-agent manufacturing systems," in *Proc. IEEE 21st Int. Conf. on Automation Science and Engineering (CASE)*, 2025, pp. 2088–2095, doi: 10.1109/CASE58245.2025.11163802.

[8] Z. Zhao *et al.*, "A large language model-based multi-agent manufacturing system for intelligent shopfloors," *Adv. Eng. Inform.*, vol. 69, 2026, art. no. 103888; doi: 10.1016/j.aei.2025.103888.

[9] M. S. Gill, J. Vyas, A. Markaj, F. Gehlhoff, and M. Mercangöz, "Leveraging LLM agents and digital twins for fault handling in process plants," in *Proc. IEEE ETFA*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205597.

[10] B. Hofmann, N. Piechulek, S. Kreitlein, J. Franke, and P. Bruendl, "Beyond touch-based human-machine interface: Control your machines in natural language by utilizing large language models and OPC UA," *arXiv preprint arXiv:2510.11300*, 2025. [Online]. Available: https://arxiv.org/abs/2510.11300v2

[11] C. Graul, W. Żarski, D. Mikołajewski, and I. Rojek, "Ai agents in industry 4.0: AAS–OPC UA–LLM architecture as the foundation of intelligent manufacturing systems in the context of industrial enterprise implementation," *Applied Sciences*, vol. 16, no. 11, p. 5428, 2026, doi: 10.3390/app16115428.

[12] D. Schall, "Safe integration of large language models into industrial process control: A multi-agent architecture with P&ID-grounded validation," *Autonomous Intelligent Systems*, vol. 6, 2026, article no. 14; doi: 10.1007/s43684-026-00136-1.

[13] S.-C. Lin and C.-Y. Chen, "An integrated framework featuring policy-governed agentic AI for closed-loop manufacturing control with multi-source sensor–MES–ERP," *The International Journal of Advanced Manufacturing Technology*, vol. 143, pp. 6405–6428, 2026, doi: 10.1007/s00170-026-17806-2.

[14] A. Biskupovic, A. Villalonga, F. Castaño, R. E. Haber, and F. Núñez, "Agent-based supervision for service-oriented industrial cyber–physical systems," *IEEE Trans. Ind. Inform.*, vol. 21, no. 3, pp. 2719–2728, 2025, doi: 10.1109/TII.2024.3514178.

[15] Y. Xia, N. Jazdi, and M. Weyrich, "An architecture for integrating large language models with digital twins and automation systems," in *Proc. IEEE 30th Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205636.

[16] L. M. Vieira da Silva, A. Köcher, and F. Gehlhoff, "Beyond formal semantics for capabilities and skills: Model context protocol in manufacturing," in *Proc. IEEE 30th Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–4, doi: 10.1109/ETFA65518.2025.11205601.

[17] K. Köhle, D. Anicic, T. A. Runkler, K. Dorofeev, A. S. Thuluva, and R. Graf, "Virtually commissioned semantic recipes based on OPC UA skills," in *Proc. IEEE 8th Int. Conf. on Industrial Cyber-Physical Systems (ICPS)*, 2025, pp. 1–6, doi: 10.1109/ICPS65515.2025.11087832.

[18] X. Jiang *et al.*, "Agentic AI for safety-aware process monitoring and fault diagnosis: A review," *Processes*, vol. 14, no. 13, 2026, art. no. 2112; doi: 10.3390/pr14132112.

[19] Model Context Protocol contributors, "Model Context Protocol: Tools," Specification, 2026, accessed: Oct. 20, 2026. [Online]. Available: https://modelcontextprotocol.io/specification/2026-07-28/server/tools

[20] ISA, "ISA-88.00.01-2010: Batch control—part 1: Models and terminology," International Standard, 2010.

[21] ——, "ANSI/ISA-95.00.01-2025 (IEC 62264-1 mod): Enterprise-control system integration—part 1: Models and terminology," International Standard, 2025.

[22] J. Moyne, E. del Castillo, and A. M. Hurwitz, Eds., *Run-to-Run Control in Semiconductor Manufacturing*, 1st ed. Boca Raton, FL: CRC Press, 2000.

[23] R. Parasuraman, T. B. Sheridan, and C. D. Wickens, "A model for types and levels of human interaction with automation," *IEEE Trans. Syst., Man, Cybern., Part A*, vol. 30, no. 3, pp. 286–297, 2000, doi: 10.1109/3468.844354.

[24] H. Wang, C. M. Poskitt, and J. Sun, "AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents," in *Proc. IEEE/ACM 48th Int. Conf. on Software Engineering (ICSE)*, 2026, pp. 2938–2950, doi: 10.1145/3744916.3764546.

[25] E. Debenedetti *et al.*, "Defeating prompt injections by design," 2025, arXiv preprint arXiv:2503.18813. [Online]. Available: https://arxiv.org/abs/2503.18813

[26] R. Parasuraman and D. H. Manzey, "Complacency and bias in human use of automation: An attentional integration," *Hum. Factors*, vol. 52, no. 3, pp. 381–410, 2010, doi: 10.1177/0018720810376055.

[27] L. Bainbridge, "Ironies of automation," *Automatica*, vol. 19, no. 6, pp. 775–779, 1983, doi: 10.1016/0005-1098(83)90046-8.

[28] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. 16th ACM Workshop on Artificial Intelligence and Security*, 2023, pp. 79–90, doi: 10.1145/3605764.3623985.

[29] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, "Formalizing and benchmarking prompt injection attacks and defenses," in *Proc. 33rd USENIX Security Symposium (USENIX Security 24)*, 2024, pp. 1831–1847. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/liu-yupei

[30] IEC, "IEC 62443-3-3:2013: Industrial communication networks—network and system security—part 3-3: System security requirements and security levels," Standard, 2013.

[31] ——, "IEC 61511-1:2016: Functional safety—safety instrumented systems for the process industry sector," International Standard, 2016.